

The Lagrange Top: Cubic Nutation, Two-Phase Reconstruction, and Exact Closure

Source-local theorem reconstruction

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Abstract

For the positive-inertia Lagrange symmetric heavy top we close the regular reduction–reconstruction problem. The inclination is inverted by one Jacobi function, both cyclic phases are complete third-kind elliptic integrals, and the full configuration closes exactly when two normalized phase increments are rational. Pole and repeated-root degenerations are kept outside singular Euler formulas. Exact computation certifies conventions, not the continuum proof.

1 Frozen convention and reduced cubic

Let $A = I_1 = I_2 > 0$, $C = I_3 > 0$, and $\gamma > 0$. In regular z - y - z Euler angles put $u = \cos \theta$, $L = p_\phi$, and $G = p_\psi$. Our convention is

$$H = \frac{p_\theta^2}{2A} + \frac{(L - Gu)^2}{2A(1 - u^2)} + \frac{G^2}{2C} + \gamma u. \quad (1)$$

The two momenta and $E = H$ are conserved. Since $p_\theta = A\dot{\theta}$, energy gives the reduced cubic owner

$$A^2 \dot{u}^2 = P(u) := 2A \left(E - \frac{G^2}{2C} - \gamma u \right) (1 - u^2) - (L - Gu)^2. \quad (2)$$

In particular

$$P(1) = -(L - G)^2, \quad P(-1) = -(L + G)^2. \quad (3)$$

The smooth Hamiltonian flow is complete: its potential is bounded on compact $\text{SO}(3)$, and positive kinetic energy makes each energy sublevel compact.

Theorem 1 (regular reduction). *A nonsteady regular inclination has finite period if and only if its allowed component of $P \geq 0$ is bounded by two simple roots in $(-1, 1)$. In the pole-incompatible chamber $L \neq \pm G$, label the roots*

$$-1 < r_1 < r_2 < 1 < r_3, \quad P = 2A\gamma(u - r_1)(u - r_2)(u - r_3).$$

This is the complete root classification for the regular reduced motion.

Proof of the reduction. Allowed positions are exactly $P \geq 0$. At a simple turning root the time integral is locally $\int s^{-1/2} ds$, while a double root gives $\int s^{-1} ds$. Equation (3), the positive leading coefficient, and the cubic sign pattern give the stated ordering. This also separates finite nutation from a separatrix.